

Algebraic Topology

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Contents

1	Complexes	2
1.1	Simplex and Simplicial complex	2
1.1.1	Simplex	2
1.1.2	Geometric simplicial complex	3
1.1.3	Abstract simplicial complex	3
1.2	Singular complex and Δ -complex/ Δ -set	3
1.3	CW complex	3
2	Classification of topological manifolds(first encounter)	3
2.1	2-manifolds	3
2.1.1	Compact surfaces	3
2.2	3-manifolds	3
3	Homotopy	3
3.1	Basic definition	4
3.1.1	Variations of homotopy equivalence	4
3.1.2	Homotopy groups	4
3.2	Homotopy equivalence and homeomorphism	4
3.3	Homotopy invariant properties	4
3.3.1	Homotopy as a category	5
3.4	Calculation of homotopy groups: Seifert-Van-Kampen	5
4	Covering maps	5
4.1	Universal coverings	5
5	Homology theory(Simplicial, Singular and CW)	5
5.1	Simplicial homology	5
5.2	Singular homology	5
5.2.1	Boundary operator	6
5.3	Singular homology groups	7
5.3.1	Definition	7
5.3.2	Group properties of singular homology groups	8
5.3.3	Functorial properties of singular homology groups 1	8
5.3.4	Functorial property: Mayer-Vietoris	9
5.4	Singular homology group and fundamental group	9
5.5	Application: degree theory	9
6	Cohomology theory	9

1 Complexes

The relation between different complexes is illustrated in:

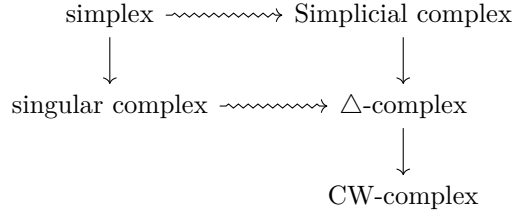


Figure 1: Complexes

1.1 Simplex and Simplicial complex

1.1.1 Simplex

Definition 1.1: Simplex

Let $v_0, v_1, \dots, v_p \in \mathbb{R}^n$ be $p + 1$ affine independent^a points in $\mathbb{R}^n$. A **geometric p -simplex** is a subset of $\mathbb{R}^n$, denoted $[v_0, v_1, \dots, v_p]$,

$$[v_0, v_1, \dots, v_p] := \left\{ \sum_{i=0}^p t_i v_i : \forall i, 0 \leq t_i \leq 1, \sum_{i=0}^p t_i = 1 \right\}$$

If write $S = \{v_0, v_1, \dots, v_p\}$, then denote $\Delta_p^S := [v_0, v_1, \dots, v_p]$. The **standard p -simplex**, denoted Δ_p , is defined as

$$\Delta_p := [e_0, e_1, \dots, e_p]$$

where e_i s are the standard basis vectors for $\mathbb{R}^p$.

^aThis can be thought as: $v_1 - v_0, \dots, v_p - v_0$ are linearly independent.

Remark Each simplex Δ_p^S can be decomposed into its boundary and interior (denoted $\mathring{\Delta}_p^S$, called the **open p -simplex**), $\Delta_p^S = \partial \Delta_p^S \cup \mathring{\Delta}_p^S$.

This process can be continued on each $p - 1$ simplex of the boundary $\partial \Delta_p^S$. Then, we get the Decomposition of a simplex:

Proposition 1.1: Equivalence of affine and linear

Proposition 1.2: Existence and uniqueness of a simplex map

Let $\Delta_p^S \subseteq \mathbb{R}^n$ be a simplex with $S = \{v_0, \dots, v_p\}$. Then, for any $p + 1$ points $w_0, \dots, w_p \in \mathbb{R}^m$, $\exists$ an affine map $F : \mathbb{R}^n \rightarrow \mathbb{R}^m$ such that $\exists!$ map $f : \Delta_p^S \rightarrow \mathbb{R}^m$ with $F|_{\Delta_p^S} = f$ and $v_i \mapsto w_i$ for all i .

Proof: First consider the translation map $x \mapsto x - v_0$ and $y \mapsto y - w_0$. Since Δ_p^S is a simplex, S is affinely independent $\Leftrightarrow \{v_1 - v_0, \dots, v_p - v_0\}$ is linearly independent by Proposition 1.1. Extend this set to a basis of $\mathbb{R}^n$, $\{v_1 - v_0, \dots, v_p - v_0, u_{p+1}, \dots, u_n\}$. Then, define $L : \mathbb{R}^n \rightarrow \mathbb{R}^m$ to be the linear extension of

$$\begin{cases} v_i - v_0 \mapsto w_i - w_0 & 0 \leq i \leq p \\ u_i \mapsto 0 & i > p \end{cases}$$

To recover an affine map, consider one possible $F(x) := L(x - v_0) + w_0$. Define $f := F|_{\Delta_p^S}$. If there is another $\tilde{F}$ satisfying the $v_i \mapsto w_i$ condition, then

$$\tilde{F}\left(\sum_i t_i v_i\right) = \sum_i t_i w_i = F\left(\sum_i t_i v_i\right)$$

showing that $\tilde{F}|_{\Delta_p^S} = F|_{\Delta_p^S}$ and then the uniqueness of f . □

Definition 1.2: Morphisms between simplexes

The unique f in Proposition 1.2 is called **the affine map determined by the vertex map** $v_i \mapsto w_i$, $i = 1, \dots, k$.

This is useful to define the morphism between two (geometric) simplicial complexes.

1.1.2 Geometric simplicial complex

Simplicial complexes are more complex than simplexes. In fact, simplexes are building blocks of simplicial complexes. It could be as the following shapes: But it couldn't be the form like: To give simplicial complex a definition. We first need to involve all the faces of each simplicial complex. Moreover, either two simplicial complexes has no intersection, or once they intersect, they must have some part of their faces coincide. Another observation is that once the interiors of two simplicial complexes do not intersect, the 'either-or' condition is automatically satisfied. Is this true exactly? The following example gives an example against this definition:

The interior of the triangle does not intersect with the interior of the line segment. It seems that this condition fails. However, by slight modification of 'the interior of two simplicial complexes' to 'the interior of every face of two simplicial complexes', this definition works well. Still look at this example, the interior of the line segment intersects with the interior of the point on this segment—the point itself. So, it seems well and we adopt this modified condition as our definition as follows:

Definition 1.3: Simplicial complex

Remark One might ask that in a simplicial complex, some simplexes share the same faces. But in this definition, no functions or relations explicitly appear to correspond with such identification of faces. Is there a flaw in this definition? Even though the relations are not explicit, be aware that all these complexes lie in $\mathbb{R}^n$. $\mathbb{R}^n$ is an ambient space that involves such relations implicitly.

Definition 1.4: Morphisms between simplicial complexes

1.1.3 Abstract simplicial complex

1.2 Singular complex and Δ -complex/ Δ -set

1.3 CW complex

CW complex is a generalization of both manifold and simplicial complex. How to see this? Instead of gluing simplexes, it glues topological balls in different dimensions.

2 Classification of topological manifolds (first encounter)

2.1 2-manifolds

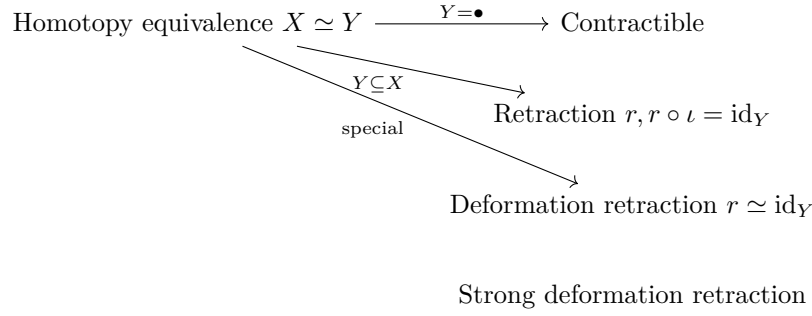
2.1.1 Compact surfaces

2.2 3-manifolds

3 Homotopy

Goals: (1) Distinguish concepts, tell their differences, (2) Be clear with examples under each concept, (3) Take it as my own language

3.1 Basic definition



Definition 3.1: Homotopy

Let X, Y be two topological spaces and $f_\nu : X \rightarrow Y$, $\nu = 0, 1$ be two continuous maps. f_0, f_1 are **homotopic**, denoted $f_0 \simeq f_1 \Leftrightarrow \exists$ a continuous map

$$F : X \times [0, 1] \rightarrow Y$$

such that $\forall \nu, \forall x \in X, F(x, \nu) = f_\nu(x)$. In this case, F is called a **homotopy** from f_0 to f_1 .

Homotopy is an equivalence relation. • Reflexivity: choose F such that $\forall \nu, F(x, \nu) = f(x)$.

• Symmetry: $f_0 \simeq f_1$ determines an F , choose $\bar{F}$ to satisfy $\bar{F}(x, \nu) = F(x, 1 - \nu)$, which is a homotopy making $f_1 \simeq f_0$.

• Transitivity: if $f \simeq g$ and $\tilde{f} \simeq \tilde{g}$ with $f, g : X \rightarrow Y$ and $\tilde{f}, \tilde{g} : Y \rightarrow Z$ being continuous maps, determining homotopy F and $\bar{F}$ respectively. Then,

Definition 3.2: Homotopy equivalence

3.1.1 Variations of homotopy equivalence

Let X be a topological space and $Y \subset X$ be its subspace. Intuitively speaking [Borsuk1931], a retraction of X is a continuous mapping $X \rightarrow Y$ that **preserves the position of all points in Y** . A deformation retraction is a mapping that

Definition 3.3: Deformation retraction

Let X be a topological space and $Y \subseteq X$ be its subspace. Let $r : X \rightarrow Y$ be the retraction of Y in X . Y is a deformation retraction of $X \Leftrightarrow r$ is homotopic to id_X

Remark $r \simeq \text{id}_Y$

3.1.2 Homotopy groups

3.2 Homotopy equivalence and homeomorphism

Homeomorphism is a special type of homotopy equivalence. In homeomorphism, if $f : X \rightarrow Y$ is a homeomorphism and $g : Y \rightarrow X$ is its inverse, then $g \circ f = \text{id}_X$ and $f \circ g = \text{id}_Y$ (here they are 'equal to', $=$, not only 'homotopy equivalent to', $\simeq$)

3.3 Homotopy invariant properties

Let X and Y be two topological spaces such that they are homotopic equivalent, i.e. $X \simeq Y$.

- Path connectivity
- Simply-connectedness
- (Singular) homology and cohomology group. The (singular) homology and cohomology groups of X and Y are isomorphic. $H_i(X) \cong H_i(Y)$ and $H^i(X) \cong H^i(Y)$

- Homotopy groups: the homotopy group of X and Y (both fundamental and higher) are isomorphic. i.e. if there is an homotopy equivalence $f : X \rightarrow Y$, then $\forall n$,

$$\pi_n(X, x_0) \cong \pi_n(Y, f(x_0))$$

3.3.1 Homotopy as a category

The topological spaces and being homotopic between them is a reminiscent of a new category: is there a category whose objects are topological spaces and whose morphisms are homotopy between two topological spaces? The answer is no, because if $f : X \rightarrow Y$ and $g : Y \rightarrow Z$ are two homotopy map, then there is no guarantee that $g \circ f$ is a homotopy.

However, if the morphisms are homotopic equivalence classes between two topological spaces $[f] := \{g : X \rightarrow Y : g \sim f\}$, then this makes a new category.

Definition 3.4: Homotopy category

The **homotopy category**, denoted **hTop**, is the category whose objects are topological spaces and whose morphisms are homotopy equivalence classes.

Remark This could be another example which reminds us that the morphisms in a category do not go to the element-level. It only describes the ability of combining two objects and does not manipulate the element directly. So, even though the equivalence classes do not look like a map, they still can be regarded as morphisms.

The well-definedness comes from the [Position of this theorem](#)

Theorem 3.1: Brouwer fixed point theorem: 2-dim case(first encounter)

Every continuous map $f : \mathbb{D}^1 \rightarrow \mathbb{D}^1$ has a fixed point.

Proof: Suppose that f has no fixed point. Then, $\forall z \in \mathbb{D}^1$, $f(z) \neq z$. Define the map

$$r(z) := \frac{z - f(z)}{\|z - f(z)\|}$$

which is a map $r : \mathbb{D}^1 \rightarrow \partial\mathbb{D}^1 \cong_{\mathbf{Top}} \mathbb{S}^1$.

Consider the restriction $r|_{\partial\mathbb{D}^1} : \partial\mathbb{D}^1 \rightarrow \partial\mathbb{D}^1$. $r|_{\partial\mathbb{D}^1}$ is homotopic to $\text{id}_{\partial\mathbb{D}^1}$, $r|_{\partial\mathbb{D}^1} = r \circ \iota_{\partial\mathbb{D}^1} \simeq \text{id}_{\partial\mathbb{D}^1}$. Because we can define $H : \partial\mathbb{D}^1 \times I \rightarrow \partial\mathbb{D}^1$ to be

$$H(x, t) := (1 - t) \cdot \text{id}_{\partial\mathbb{D}^1}(x) + t \cdot r|_{\partial\mathbb{D}^1}(x)$$

Since $\iota_{\partial\mathbb{D}^1}$ is a null homotopy. So is $r \circ \iota_{\partial\mathbb{D}^1}$. But, $\text{id}_{\partial\mathbb{D}^1}$ is not a null homotopy. Contradiction. $\square$

Remark The process of r is to first plot x and $f(x)$. Then, connect them, parallel this to 0, and extend it to intersect it with the $\partial\mathbb{D}^1$.

A tendency is to think 0 as a special point and $f(0) = 0$. f is in the **Top** category. Any point is not special at all. Under a homeomorphism, any point in $\mathbb{D}^1$ can be moved to the 'center'.

Why disk?

3.4 Calculation of homotopy groups: Seifert-Van-Kampen

4 Covering maps

4.1 Universal coverings

5 Homology theory(Simplicial, Singular and CW)

5.1 Simplicial homology

5.2 Singular homology

To make triangle decomposition in a topological space X , it would be convenient to 'embeds' a geometric(more specifically, a standard) p -simplex into X . So,

Definition 5.1: Singular p -simplex

Let X be a topological space. A **singular p -simplex** is a continuous map $\sigma : \Delta_p \rightarrow X$.

Remark The word 'singular' here implies that σ might not be a nice embedding. The image might have 'singularities' that it does not look like a simplex at all [hatcher].

Basically, the image of σ is the desired simplex.

Example A singular 0-simplex is a point.
A singular 1-simplex is a path in X .

Definition 5.2: Singular chain group

The **singular chain group of X in degree p** , denoted by $S_p(X, \mathbb{Z})$, is the *free abelian group* generated by the basis {singular p -simplex in X , σ }

Remark The operation in $S_p(X, \mathbb{Z})$ is addition. The intuition of being free abelian groups is: those standard p -simplices (or singular p -simplices) are the fundamental building blocks of more complex objects. The 'addition' here corresponds to 'splicing', and we will see how to deal with the boundary nicely when splicing in the definition of boundary operators.

More explicitly,

$$S_p(X, \mathbb{Z}) = \mathbb{Z}[\text{Hom}_{\mathbf{Top}}(\Delta_p, X)]$$

The following map is to embed a standard simplex into a simplex with one-dimension higher as a face of the higher dimensional simplex. $\forall i$, the i -th face map in Δ_p is defined to be $F_{i,p} : \Delta_{p-1} \rightarrow \Delta_p$ by

$$f_j \mapsto \begin{cases} e_j & j < i \\ e_{j+1} & j \geq i \end{cases}$$

where f_j s are basis of Δ_{p-1} and e_i s are basis of Δ_p .

So, each $F_{i,p}$ is a topological embedding with image $F_{i,p}(\Delta_{p-1}) = [e_0, \dots, \hat{e}_i, \dots, e_p]$, the face opposite e_i . In other words, $F_{i,p}$ is the map for searching and anchoring Δ_{p-1} to a certain part of the boundary of Δ_p .

Definition 5.3: Boundary of singular p -simplex

The **boundary of singular p -simplex $\sigma : \Delta_p \rightarrow X$** , denoted $\partial_p \sigma$, is the map

$$\partial_p \sigma := \sum_{i=0}^p (-1)^i \sigma \circ F_{i,p}$$

5.2.1 Boundary operator

Since there is a notion on every singular p -simplices, and $S_p(X, \mathbb{Z})$ is generated by all singular p -simplices σ . So, the 'boundary' can be extended to the whole singular chain group $S_p(X, \mathbb{Z})$ as an operator:

Definition 5.4: Boundary operator

For X a topological space and its singular chain group in consecutive degrees p and $p-1$, the **boundary operator** between them, denoted ∂_p , is defined to be the map:

$$\partial_p : S_p(X, \mathbb{Z}) \rightarrow S_{p-1}(X, \mathbb{Z}) \quad \sigma \mapsto \partial_p \sigma := \sum_{i=0}^p (-1)^i \sigma \circ F_{i,p}$$

Boundary operator has a very important property:

Theorem 5.1: Nilpotency of boundary operator

Let $\sigma \in C_p(X)$. Then $\partial_{p-1}(\partial_p(\sigma)) = 0$.

Proof:

$$\begin{aligned}
& \partial_{p-1}(\partial_p(\sigma)) \\
&= \sum_{j=0}^{p-1} (-1)^j \partial_p(\sigma) \circ F_{j,p-1} \\
&= \sum_{j=0}^{p-1} (-1)^j \left(\sum_{i=0}^p (-1)^i \sigma \circ F_{i,p} \right) \circ F_{j,p-1} \\
&= \sum_{j=0}^{p-1} \sum_{i=0}^p (-1)^{i+j} \sigma \circ F_{i,p} \circ F_{j,p-1} \\
&= \sum_{0 \leq i \leq j \leq p-1} (-1)^{i+j} \sigma \circ F_{j,p-1} \circ F_{j,p-1} + \sum_{0 \leq j < i \leq p} (-1)^{i+j} \sigma \circ F_{i,p} \circ F_{j,p-1} \\
&= \sum_{0 \leq i \leq j \leq p-1} (-1)^{i+j} \sigma \circ F_{j,p-1} \circ F_{j,p-1} + \sum_{0 \leq j < i \leq p} (-1)^{i+j} \sigma \circ F_{j,p} \circ F_{i-1,p-1} \quad j < i \leq p \Rightarrow j \leq i-1 \leq p-1 \\
&= \sum_{0 \leq i \leq j \leq p-1} (-1)^{i+j} \sigma \circ F_{j,p-1} \circ F_{j,p-1} + \sum_{0 \leq j \leq k \leq p-1} (-1)^{k+j+1} \sigma \circ F_{j,p} \circ F_{k,p-1} \quad (k := i-1) \\
&= 0
\end{aligned}$$

□

Corollary 5.1: Exactness of singular chain groups

The collection of all singular chain groups of X , $\{S_p(X, \mathbb{Z})\}_p$, together with the boundary operators forms an exact sequence. More explicitly, there is a long exact sequence

$$\cdots \xrightarrow{\partial_{p+2}} S_{p+1}(X, \mathbb{Z}) \xrightarrow{\partial_{p+1}} S_p(X, \mathbb{Z}) \xrightarrow{\partial_p} S_{p-1}(X, \mathbb{Z}) \xrightarrow{\partial_{p-1}} \cdots$$

Proof: The theorem 5.1 implies that $\text{im } \partial_p \subseteq \ker \partial_{p-1}$. □

This theorem shows that $S_\bullet^\mathbb{Z} : \mathbf{Top} \rightarrow \mathbf{Comp}(\mathbf{AbGp})$ is a functor. The objects of $\mathbf{Top}$, topological spaces are mapped to the chain complex of abelian groups, while the continuous maps $f : X \rightarrow Y$ between two topological spaces are mapped to a morphism of two complexes $S_\bullet^\mathbb{Z}(X) \rightarrow S_\bullet^\mathbb{Z}(Y)$.

5.3 Singular homology groups

5.3.1 Definition

For each p , the singular homology group of X at p , denoted $H_p^{\text{sing}}(X)$, is defined as

$$H_p^{\text{sing}}(X) := \ker \partial_p / \text{im } \partial_{p+1}$$

By the way from general homology definition, $H_\bullet$ depends on the functor

$$S_\bullet : \mathbf{Top} \rightarrow \mathbf{Comp}(\mathbf{AbGp})$$

This shows how to rephrase $H_p^{\text{sing}}(X)$ in the general language of H_p :

$$H_p^{\text{sing}}(X) = H_p(S_\bullet(X))$$

An illustration is:

$$\begin{array}{lcl}
\text{Functors :} & \mathbf{Top} & \xrightarrow{S_\bullet} \mathbf{Comp}(\mathbf{AbGp}) \xrightarrow{H_p} \mathbf{AbGp} \\
& & \text{with a curved arrow } H_p^{\text{sing}} \text{ from } \mathbf{Top} \text{ to } \mathbf{AbGp} \\
\text{Objects :} & X & \rightsquigarrow S_\bullet(X) \rightsquigarrow H_p(S_\bullet(X))
\end{array}$$

5.3.2 Group properties of singular homology groups

Theorem 5.2: Decomposition of homology groups

Let X be a topological space and $\{X_\alpha\}_\alpha$ be its collection of connected components. Then, there is a group homomorphism

$$\bigoplus_{\alpha} H_p^{\text{sing}}(X_\alpha) \rightarrow H_p^{\text{sing}}(X)$$

Corollary 5.2: Homology of discrete space

Let X be a discrete space, then $H_0^{\text{sing}}(X)$ is a free abelian group with 1-generator. For $p > 0$, $H_p^{\text{sing}}(X) = 0$.

Proof:

□

Corollary 5.3: Homology of

5.3.3 Functorial properties of singular homology groups 1

From the category of topological spaces $\mathbf{Top}$ to the category of abelian groups. Functoriality of singular homology groups

Theorem 5.3: Functoriality of singular homology groups

Let X, Y be topological spaces with a continuous map $f : X \rightarrow Y$. The induced map $f_p : H_p(X) \rightarrow H_p(Y)$ is a group homomorphism, $f_* : H_\bullet(X) \rightarrow H_\bullet(Y)$

Singular homology group is a topological invariant.

More specifically, this 'functor' could be refined, from the category of homotopy category $\mathbf{hTop}$ to the category of abelian groups $\mathbf{AbGp}$. Recall that in the category $\mathbf{hTop}$, objects are still topological spaces and the morphisms are equivalence classes of homotopic maps.

The next theorem implies how this 'functor' manipulates the morphisms:

Theorem 5.4

Let X, Y be topological spaces and $f, g : X \rightarrow Y$ are continuous and homotopic ($f \simeq g$). Then the two maps induce the same homomorphism $f_* = g_* : H_n(X) \rightarrow H_n(Y)$

Is this kind of singular homology theory can be uniformed in the language of derived functors?

5.3.4 Functorial property: Mayer-Vietoris

Theorem 5.5: Mayer-Vietoris theorem

Let X be a topological space and U, V be its open sets such that $X = U \cup V$ and U, V are not necessarily path-connected. Then, the commutative diagram

$$\begin{array}{ccc}
 & U & \\
 i \nearrow & & \searrow k \\
 U \cap V & & U \cup V =: X \\
 j \searrow & & \nearrow l \\
 & V &
 \end{array}
 \qquad
 \begin{array}{ccc}
 & H_p^{\text{sing}}(U) & \\
 i \nearrow & & \searrow k \\
 H_p^{\text{sing}}(U \cap V) & & H_p^{\text{sing}}(X) \\
 j \searrow & & \nearrow l \\
 & H_p^{\text{sing}}(V) &
 \end{array}$$

5.4 Singular homology group and fundamental group

Theorem 5.6: Hurwitz theorem

Let X be a connected space. For every $p \in X$, there is a surjective group homomorphism

$$\gamma : \pi_1(X, p) \rightarrow H_1^{\text{sing}}(X)$$

Then, $H_1^{\text{sing}}(X) \cong \pi_1(X, p) / \ker \gamma$.

Proof: We are going to show the followings:

- (1) γ is a surjective homomorphism.
- (2) $\ker \gamma$ is the commutator subgroup of $\pi_1(X, p)$. So, $H_1^{\text{sing}}(X)$ is the abelianization of $\pi_1(X, p)$.

□

5.5 Application: degree theory

6 Cohomology theory